

Equation word problems

Unit 5 · Algebra applicationadvanced · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5, Volume A, Unit 5 + Modern Education, Volume 5, Unit 12

Core Traps:□ T3 Text→Algebraic Translation + T7 Reverse Thinking of Equations

SSPA Related: ● **High Frequency** Compulsory application questions in Paper 2, accounting for about 15-20% of Paper 1

Prerequisite knowledge:Class 14 (Knowledge of algebraic expressions · One-step equations · Principle of balance)

Class objectives:① One-step/two-step equation application questions ② "A is more/less than B" comparison class ③ Perimeter area + equation synthesis ④ "A is n times more than B" translation

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (write out complete process)
1	Solve the equation: $x + 18 = 45$	Basic	
2	Solve the equation: $x - 23 = 17$	Basic	
3	Solve the equation: $6x = 54$	Basic	
4	Solve the equation: $x \div 8 = 12$	Advanced	
5	Use algebraic expressions: "4 times a number plus 7" and "A is 5 more than B" (assuming $A = B + ?$)	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: One-step equation word problems ● SSPA

One-step equation application problem solving framework:

- ① **Let x** |||SEP|||: Let x be the required unknown quantity (usually let x be the quantity such as "how many are there originally", "how many are there in total") Translation |||SEP|||: Translate the relationship described in words into an equation (be careful T3: sequence!)
- ② **Solve x**: Use reverse operation to find x in one step (be careful with T7: addition changes to subtraction, subtraction changes to addition, multiplication changes to division, division changes to multiplication)
- ③ **Answer Sentence** |||SEP|||: Write "Answer: _____ (unit)." - If you do not write an answer sentence in the submission test, step points will be deducted!: Use reverse operation to find x in one step (be careful with T7: addition changes to subtraction, subtraction changes to addition, multiplication changes to division, division changes to multiplication)
- ④ **answer**: Write "Answer: _____ (Unit)." - Failure to write an answer sentence will result in step points being deducted!

❑ Trap detonation example (T3: Translate "A is twice as much as B")

The older brother has twice as many stickers as the younger brother, 7 stickers. If the younger brother has x pieces, use an algebraic expression to express the number of older brothers.

✗ **T3 common errors (reversed translation order)**

$$\begin{aligned} \text{Brother} &= 2(x + 7) \\ \text{or brother} &= x + 7 \times 2 \end{aligned}$$

✓ **Correct translation**

$$\text{Brother} = 2x + 7$$

"Twice as much as younger brother" = $2x$, "7 more" = $+7 \rightarrow 2x + 7$. Note: "Many 7" should be added at the end!

"The younger brother's 2 times is more than 7" → First multiply by 2 and then add 7 = $2x+7$. It's not like adding 7 and then multiplying by 2!

 **Tip: "Suppose x must be accurate, and the translation must be correct word for word; multiples must be added and subtracted first, and then the inverse solution of the equation must be listed."**

 **The most frequent error: "A is 3 times more than B 5" → written as $3(B+5)$ instead of $3B+5$. Remember: multiply first, then subtract!**

 **The second most frequent error: Forgetting to write "Suppose x = _____" in the application question directly. Points will be deducted for the step if you submit it to the sub-test!**

Knowledge Point - Worked Examples (let $x \rightarrow$ column equation $\rightarrow$ solution $\rightarrow$ answer)

#	Question	Difficulty	Working Space
Example 1	Xiao Ming had some candies and after eating 9, he had 16 left. How many pills were there originally?		
Example 2	There are 12 eggs in a dozen eggs. If you buy x dozen, there are 60 in total. Find x.		

Knowledge point 2: Two-step equation word problems **SSPA must take the exam**

Characteristics and solutions of two-step equations:

- ① The equation contains two operations, for example: $2x + 5 = 25$ or $3x - 4 = 20$
- ② **Solution sequence |||SEP|||: Process "addition and subtraction" (peripheral) first, and then "multiplication and division" (inner layer)**
- ③ For example, $2x + 5 = 25$: In the first step, subtract 5 from both sides $\rightarrow 2x = 20$; in the second step, divide both sides by 2 $\rightarrow x = 10$
- ④ For example, $3x - 4 = 20$: In the first step, add 4 to both sides $\rightarrow 3x = 24$; in the second step, divide both sides by 3 $\rightarrow x = 8$
- ④ For example, $3x - 4 = 20$: In the first step, add 4 to both sides $\rightarrow 3x = 24$; in the second step, divide both sides by 3 $\rightarrow x = 8$

①

**Processing
additions and
subtractions**

②

**Handle
multiplication
and division**

③

Find x
Get the value of x

④

Check calculation
Check the answers layer
by layer

First eliminate the peripheral +/ -

Then eliminate the $\times/\div$ in the inner layer

example

Example 3: Xiao Ming bought 4 identical books, paid \$100 and got back \$16. How much is each book? (Suppose each book \$x $\rightarrow 4x + 16 = 100$)

example

Example 4: 3 times a number minus 8 equals 37. Find this number. (let this number be x)

Knowledge Point 2 Synchronization practice (must write a two-step process)

#	Question	Difficulty	Working Space
6	5 boxes of the same chocolates cost \$85. The price was later increased by \$3 per box. How much does each box cost after the price increase? (Suppose the original price is \$x per box)		
7	Xiao Ming is n years old this year. 5 years later, 3 times his age is 48. Find n.		
8	Five times a number plus 12 equals 47. Find this number.		
9	My sister had stickers, and after giving half to her friend, she gave 3 more stickers, and finally she was left with 7 stickers. How many pictures does your sister originally have? (Suppose there are x original pictures $\rightarrow \frac{x}{2} - 3 = 7$)		
10	A rectangle whose length is twice its breadth is 3 cm more. If the width is w cm and the length is 17 cm. Find w.		

Knowledge point 3: Comparative equations - "A is more/less than B" SSPA Advanced

T3 Trap key point - translation of comparative relationship:

- ① "A is k more than B" $\rightarrow A = B + k$ or $A - B = k$
- ② "A is k less than B" $\rightarrow A = B - k$ or $B - A = k$
- ③ is n times more than B by k" $\rightarrow A = nB$
- ④ "A is n times less than B by k" $\rightarrow A = nB + k$ **Key |||SEP|||:** What follows the word "ratio" is the base quantity (usually set to x), and what precedes the word "is" is the comparison result $\rightarrow A = nB - k$
- ⑤ **key:** What follows the word "ratio" is the base quantity (usually set to x), and what precedes the word "is" is the comparison result.

example

Example 5: A has \$150, and B has \$38 less than A. How many yuan does B have?
(Suppose B has \$x $\rightarrow A = x + 38$ or $x = 150 - 38$)

example

Example 6: A's age is 3 times that of B. The sum of the ages of A and B is 48 years. Find the ages of A and B. (Let B = x)

□ T3+T7 double trap comparison question

There are 2 times as many apples as 5 oranges. Apple has 35. How many oranges are there?

✗ T3 translation error + T7 reverse error

Let orange = x

$$35 = x \div 2 - 5$$

"An apple is 2 times less than an orange and is 5 less"

$\rightarrow$ Apple = $2x - 5$. Not $x \div 2 - 5$! And when solving inversely, you must first +5 and then $\div 2$

✓ Correct solution

$$35 = 2x - 5$$

$$2x = 40$$

$$x = 20$$

x oranges $\rightarrow$ apples = $2x - 5 = 35 \rightarrow 2x = 40 \rightarrow x = 20$.

Calculation: $2 \times 20 - 5 = 35$ ✓

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
11	There are 18 more Chinese books than English books. There are 52 Chinese books. How many English books are there? (Suppose there are x English books)		
12	A is 4 times larger than B. A and B have a total of 75. Find the value of A and B.		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
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13	There are x pieces in a box of sugar. After eating 14 pills, I have 26 pills left. How many pills were there originally?		
14	8 identical oranges weigh a total of 2.4 kg. How many kilograms does each orange weigh? (Let each weigh x kg)		
15	Dad is 42 years old, 3 years older than mom. How old is your mother this year? (Assume mother is y years old)		
16	Seven times a number is 91. Find this number.		
17	Xiao Ming has \$ x , and after spending \$28, he is left with \$35. How much money did he originally have?		

Advanced layer (total 5 questions, choose do)

#	Question	Difficulty	Working Space
18	Three times a number plus 14 equals 50. Find this number.		
19	Bought 5 identical school bags, paid \$500 and got \$75 back. How much is each school bag?		
20	A and B have a total of \$280. A is 3 times as big as B. Find out how much yuan each of them has.		
21	Add 8 to a number and multiply it by 3. The result is 54. Find this number. (Let this number be $x \rightarrow 3(x+8) = 54$)		
22	The younger brother's age is 2 years younger than his older brother's SEP . If the older brother is 16, how old is the younger brother? If the younger brother is 5 years old, how old is the older brother? (Second question of solving equations) $\frac{1}{2}$ 2 years younger. If the older brother is 16, how old is the younger brother? If the younger brother is 5 years old, how old is the older brother? (Second question of solving equations)		

challenge layer (total 5 questions, choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
23	The sum of three consecutive numbers is 84. Find these three numbers. (Let the smallest one be x)		
24	A bottle of soda originally had some. After drinking half of it, pour in another 200 mL, now you have 650 mL. How many mL was there originally? (make a two-step equation)		
25	Cut a rope into two pieces. The longer section is 3 times more than the short one, 2 m. If the total length is 26 m, find the length of each of the		

	two segments in m.		
26	The page numbers of a book. I read $\frac{1}{3}$ on the first day, and the remaining $\frac{1}{2}$ on the second day. There are still 40 pages left to read. How many pages are there in the book? (Assume x pages of the book)		
27	A and B have a total of \$450. After A spends \$50, A has 3 times as much money as B. Find the original dollars of A and B. (Suppose B has \$x)		

Knowledge Point 4: Comprehensive Application - Perimeter/Area + Equation

SSPA required test

Geometry + Equation problem-solving framework (common in sub-test paper 2):

- ① **Drawing symbols |||SEP|||: Draw the figure first, label the known length and unknown numbers** Recall formulas |||SEP|||: Perimeter/area formula (rectangular perimeter = $2(L+W)$, area = $L \times W$, etc.)
- ② **Let x |||SEP|||: Usually, let the shortest side or base side be x** Substitute the formula $\rightarrow$ column equation |||SEP|||
- ③ **Let x:** Usually let the shortest side or base side be x
- ④ **Substituting formula $\rightarrow$ column equation:** Substituting the algebraic expression into the formula
- ⑤ **Solve the equation $\rightarrow$ Substitute back to find the sides:** After finding x, substitute back to find all side lengths
- ⑥ **Answer sentence unit**

example

Example 7: The length of a rectangle is twice its width. The perimeter is 48 cm. Seek length and breadth. (Suppose width = w cm $\rightarrow$ length = $2w \rightarrow 2(2w+w) = 48$)

example

Example 8: The length of a rectangle is 5 cm more than its width. The area is 84 cm^2 . If the width is w cm: (a) Express the length algebraically. (b) Here is the equation. (c) Find length and breadth.

 **The most common mistake in geometry word problems: forgetting to multiply the perimeter formula by 2! Perimeter = (length + width) $\times$ 2, not length + width!**

application question practice (let unknown $\rightarrow$ equation $\rightarrow$ solution $\rightarrow$ answer sentence, advanced requires drawing assistance)

28	The side length of the square is s cm. The perimeter is 36 cm. Find s .		
29	The width of the rectangle is 5 cm and the length is 2 times the width. Find the perimeter. (It can be solved without an equation, but try the equation: let width = w)		
30	The length of a rectangle is 3 times its width. The perimeter is 64 cm. Seek length and breadth.		
31	The length of the rectangle is 4 cm more than its width. The perimeter is 40 cm. Seek length and breadth.		
32	The sum of the sides of the triangle is 30 cm. The second side is 2 times as long as the first side and the third side is 3 cm longer than the first side. Find the length of each of the three sides.		
33	The length of the rectangle is 12 cm and the width is y cm. (a) Express the perimeter and area algebraically. (b) If the perimeter is 40 cm, find y . (c) Find the area using the value of y .		
34	The upper base of a trapezoid is a cm, the lower base is 2 times the upper base, and the height is 5 cm. The area is 45 cm^2 . (Trapezoid area = (upper base + lower base) $\times$ height $\div$ 2) Find the upper base a .		
35	A rectangular garden is 5 m longer than twice as long as it is wide. If the perimeter is 70 m, find the length, breadth and area of the garden.		

IV. Ultimate challenge area

#	Question	Difficulty	Working Space
 1	A, B, and C have a total of \$540. B is twice as large as A, and C is three times as large as B. Find out how much each of the three people has. (Suppose A has \$ x)		
 2	The length of a rectangle is twice its width. If the length and width each increase by 3 cm, the area increases by 57 cm^2 . Find the original length and breadth. (Assume original width = x cm)		
 3	A two-digit number, the tens digit is 3, and the ones digit is x . (a) Express this number algebraically. (b) If the tens and ones digits are reversed, what is the new number? (c) If the new number is 27 greater than the original number, find x .		

V. Class afterhomework

Basic must-doquestions (total 5 questions, mustlet $x \rightarrow$ equation $\rightarrow$ solution $\rightarrow$ answer)

#	Question	Difficulty	Working Space
H1	Xiaomei has x yuan, and after spending \$45 on books, she has \$38 left. How much money did she originally have?		
H2	Nine times a number is 108. Find this number.		
H3	Three times a number plus 11 equals 47. Find this number.		
H4	A and B have a total of \$156. A is 5 times larger than B. Find out how much yuan each of them has.		
H5	The length of a rectangle is 3 times its width. The perimeter is 56 cm. Seek length and breadth.		

Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	There are 88 Chinese and English books in total. There are 12 more Chinese books than English books. How many copies of each type of book are there? (Suppose there are x English books)		
H7	The sum of the sides of the triangle is 45 cm. The second side is 2 times the first side and the third side is 5 cm less than the second side. Find the length of each of the three sides.		
H8	Five years later, the father is three times as old as the son. Now the father is 40 years old, how old is the son? (Suppose the son is now x years old $\rightarrow 40+5 = 3(x+5)$)		

VI. The Lessoncorecommon errorssummary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

 Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve 🧙 basic questions independently (100% correct)

☐ Challenge 🌿 Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	T3: "A is n times more k than B" translation error SEP : written as $n(B+k)$ instead of $nB+k$: Written as $n(B+k)$ instead of $nB+k$	Multiples first: multiply n first, then add k $\rightarrow$ $nB+k$. Only "A is k times more than B" is $n(B+k)$
2	T3: Reverse direction of comparison SEP : "A is 5 more than B" $\rightarrow$ Suppose $A=x$, $B=x+5$: "A is 5 more than B" $\rightarrow$ Suppose $A=x$, $B=x+5$	After the word "bi" is the benchmark! "A is 5 more than B" $\rightarrow$ $A = B+5$, usually assuming $B=x$
3	T7: Error in solving the two-step equation SEP : $2x+5=25$ first $\div 2$ and then -5 : $2x+5=25$ first $\div 2$ and then -5	Process addition and subtraction first (peripheral), then multiplication and division (inner layer): first -5 and then $\div 2$
4	T7: Reverse thinking error SEP : Solution $x \div 3+2=10 \rightarrow x \div 3=10+2$: Solve $x \div 3+2=10 \rightarrow x \div 3=10+2$	Reverse: $x \div 3+2=10 \rightarrow x \div 3=10-2=8 \rightarrow x=8 \times 3=24$
5	Forgot the formula or used the wrong formula in geometry questions SEP : Perimeter = length + width: Perimeter = length + width	Perimeter of rectangle = $2(L+W)$, perimeter of square = $4s$. Be sure to remember the brackets!
6	Suppose x and then omit the algebraic expression of other quantities SEP : Suppose only x and do not express other relationships: Only set x, do not express other relationships	After setting x, immediately use x to write the algebraic expressions of all relevant quantities
7	Missing sentences, missing units, and missing calculations in application questions	Full format: Let $x \rightarrow$ Equation $\rightarrow$ Solve $\rightarrow$ Check $\rightarrow$ Answer: _____ (unit)

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